Annexure – III

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

STATEMENT OF EXPENDITURE IN RESPECT OF MAJOR RESEARCH PROJECT

1. Name of Principal Investigator: **Dr. Ruth Shanthakumari T**
2. Dept. of Principal Investigator: Department of Zoology
3. University/College: St. Philomena's College, Mysore
4. UGC approval Letter No. and Date: **F.No. 42-615/2013(SR) dated 25th March 2013**
5. Title of the Research Project: **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*"**
6. Effective date of starting the project: 07/05/2013
7. **a.** Period of Expenditure: From **01-07-2013** to **31-03-2017 (3 years+1 year extension)**

b. Details of Expenditure:

Sl No.	Item	Amount Approved (Rs.)	Amount Sanctioned (Rs)	Expenditure Incurred (Rs.)	Additional amount of 10% of the grant spent by the institute (Rs.)	Final expenditure incurred (Rs.)
A.	**Non-Recurring**					
1	Books & Journals	Nil	Nil	nil	Nil	Nil
2	Equipment	Nil	Nil	nil	Nil	Nil
B.	**Recurring**					
1.	Honorarium	Nil	Nil	nil	Nil	Nil
2.	Project fellow	5,08,903/-	4,58,012	4,57,934/-	50,666/-	5,08,600/-
3.	Chemicals/ Glassware/ Consumable	1,50,000/-	1, 35, 000/-	1, 34, 981/-	15,021/-	1,50,002/-
4.	Hiring Services	Nil	Nil	nil	Nil	Nil
5.	Contingency	50,000/-	45,000/-	44,977/-	5,023/-	50,000/-
6.	Travel/Field work	35,000/-	31,500/-	31,473/-	3,530/-	35,003/-
7.	Special Need	Nil	Nil	Nil	Nil	Nil
8.	Overhead Charges	72,800/-	72,800/-	69,988/-	-	69,988/-
	Total	**8, 16, 703/-**	**7, 42, 312/-**	**7, 39, 353/-**	**74,240/-**	**8,13,593/-**

Balance recurring charges (10% of the total grant) of Rs. 74,391/- (Honorarium: 50,891/-, Chemical/Glassware/Consumable: 15,000/-, Travel/Field work: 3,500/- and Contingency: 5000/-) needs to be released by the UGC-major research project, New Delhi.

C. Staff - I

1. Mr. Goutham S

Date of Appointment: 07/05/2013

Sl. No	Items	From	To	Amount approved (Rs.)	Amount Sanctioned (Rs)	Expenditure incurred (Rs.)
1	**Honorarium to PI** (Retired Teachers) @ Rs. 18,000/-p.m.	-	-	-	-	-
2	***Project fellow:** @Rs.14,000/- p.m. for initial 2 years and Rs.16,000/- p.m. from third year onwards.	07/05/2013	06/04/2016	5,08, 903/-	**4,58, 012/-**	**5,08,600/-**
3	**HRA** @ 20%	-	-	-	-	-

1. It is certified that the appointment(s) have been made in accordance with the terms and conditions laid down by the Commission.
2. If as a result of check or audit objection some irregularly is noticed at later date, action will be taken to refund, adjust or regularize the objected amounts.
3. Payment @ revised rates shall be made with arrears on the availability of additional funds.
4. It is certified that the grant of Rs. **7, 42, 312/-** (Rupees Seven lakh Forty Two thousand Three hundred and Twelve only) received from the University Grants Commission under the scheme of support for Major Research Project entitled **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*"** vide UGC letter No. F. 42-615/2013(SR) dated 25th March 2013 **(I-grant)** and dated 29th Aug 2015 **(II-grant). Rupees Seven lakh Thirty Nine thousand Three hundred and Fifty Three (7, 39, 353/-)** has been utilized for the purpose for which it was sanctioned and in accordance with the terms and conditions laid down by the University Grants Commission.
5. Amount of **Rupees Seventy Four thousand Two hundred and Forty 74,240/- (Honorarium: 50,666/-,** Chemical/Glassware/Consumable: 15,021/-, Travel/Field work: 3,530/- and Contingency: 5,023/-) under recurring charges has been spent by the institute in advance, and will be reimbursed by the institute once it is released by the UGC-MRP.

PRINCIPAL INVESTIGATOR
Dr. T. RUTH SHANTHAKUMARI
Principal Investigator-MRP
Dept. of Zoology
St. Philomena's College, Mysore-570 015

PRINCIPAL
Principal
St. Philomena's College
MYSORE-570 015

Annexure – IV

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

STATEMENT OF EXPENDITURE INCURRED ON CHEMICAL/GLASSWARE/CONSUMABLES

Name of the Principal Investigator: Dr. Ruth Shantha Kumari T

Sl. No.	Description of the article	Bill No. and Date	Amount in Rs.	Name of the firm
1.	Chemicals purchased for the UGC-Major research project	No. RCM0833/13-14, date 08.10.2013	44,139-00	Raghu Chemicals, K.T. Street, Mysore.
2.	SDS-PAGE Protein electrophoresis unit purchased for the UGC-Major research project	No. 1131, date 24.02.2014	16,316-00	Sree Maruthi Scientific Works, Bangalore.
3.	Glassware purchased for the UGC-Major research project	No. 15171, date 04.08.2014	13,836-00	Nidhi Agencies, Siddhartha Nagar, Mysore.
4.	Chemicals (consumables) purchased for the UGC-Major research project	No. 383, date 23.12.2015	2,020-00	Sri Ragavendra Enterprises.
5.	Chemicals purchased for the UGC-Major research project	No. 17070, date 07.01.2016	18,940-00	Nidhi Agencies, Siddhartha Nagar, Mysore.
6.	Chemicals (glassware) purchased for the UGC-Major research project	No. 17068, date 07.01.2016	19,823-00	Nidhi Agencies, Siddhartha Nagar, Mysore.
7.	Chemicals (consumables) purchased for the UGC-Major research project	No. 17069, date 07.01.2016	19,907-00	Nidhi Agencies, Siddhartha Nagar, Mysore
8.	Glassware for the UGC-Major research project	No. 72, 84, 88, 91. Date 15.02.2016	15,021-00	Bhuvaneshwari Unit, Mysore.
		Total	**1,50,002-00**	
Rupees One Lakh Fifty Thousand and Two Rupees				

Certified that the above expenditure is in accordance with the UGC norms for Major Research Projects.

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Annexure – IV

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

STATEMENT OF EXPENDITURE INCURRED ON CONTINGENCY

Name of the Principal Investigator: Dr. Ruth Shantha Kumari T

Sl. No.	Description of the article	Date	Amount in Rs.	Name of the firm
1.	Stationary charges incurred in four separate bills for the UGC-Major research project,	From 18.05.2013 to 19.01.2016	**11,467-00**	Purchased from various firms for the purpose of project.
			13,168-00	
			20,342-00	
			5,023-00	
		Total	**50,000-00**	
Rupees Fifty Thousand only				

Certified that the above expenditure is in accordance with the UGC norms for Major Research Projects.

PRINCIPAL INVESTIGATOR
Dr. T. RUTH SHANTHAKUMARI
Principal Investigator-MRP
Dept. of Zoology
St. Philomena's College, Mysore-570 015

PRINCIPAL
Principal
St. Philomena's College
MYSORE-570 015

Annexure – IV

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

STATEMENT OF EXPENDITURE INCURRED ON TRAVEL/FIELD WORK

Name of the Principal Investigator: Dr. Ruth Shantha Kumari T

BILL NO. 1

Sl. No.	Description of the article	Bill No. and Date	Amount in Rs.	Name of the firm
1.	Auto fare for the collection of specimen (garden lizard).	No. 01, date 19.05.2013	500-00	Nataraja, Auto no. KA 09 A 3914
2.	Auto fare for the collection of specimen (garden lizard).	No. 02, date 09.06.2013	500-00	Nataraja, Auto no. KA 09 A 3914
3.	Auto fare for the collection of specimen (garden lizard).	No. 03, date 23.06.2013	600-00	Nataraja, Auto no. KA 09 A 3914
4.	Auto fare for the collection of specimen (garden lizard).	No. 04, date 07.07.2013	650-00	Nataraja, Auto no. KA 09 A 3914
5.	Auto fare for the collection of specimen (garden lizard).	No. 05, date 21.07.2013	450-00	Puttaswamy, Auto no. KA 09 A 2437
6.	Auto fare for the collection of specimen (garden lizard).	No. 06, date 04.08.2013	550-00	Puttaswamy, Auto no. KA 09 A 2437
7.	Auto fare for the collection of specimen (garden lizard).	No. 07, date 25.08.2013	350-00	Nataraja, Auto no. KA 09 A 3914
8.	Auto fare for the collection of specimen (garden lizard).	No. 08, date 08.09.2013	550-00	Nataraja, Auto no. KA 09 A 3914
9.	Auto fare for the collection of specimen (garden lizard).	No. 09, date 22.09.2013	450-00	Nataraja, Auto no. KA 09 A 3914
10.	Auto fare for the collection of specimen (garden lizard).	No. 10, date 06.10.2013	400-00	Puttaswamy, Auto no. KA 09 A 2437
11.	Auto fare for the collection of specimen (garden lizard).	No. 11, date 20.10.2013	600-00	Nataraja, Auto no. KA 09 A 3914
12.	Auto fare for the collection of specimen (garden lizard).	No. 12, date 03.11.2013	550-00	Nataraja, Auto no. KA 09 A 3914
13.	Auto fare for the collection of specimen (garden lizard).	No. 13, date 17.11.2013	500-00	Nataraja, Auto no. KA 09 A 3914
14.	Auto fare for the collection of specimen (garden lizard).	No. 14, date 08.12.2013	350-00	Nataraja, Auto no. KA 09 A 3914
15.	Auto fare for the collection of specimen (garden lizard).	No. 15, date 22.12.2013	450-00	Nataraja, Auto no. KA 09 A 3914
16.	Auto fare for the collection of specimen (garden lizard).	No. 16, date 05.01.2014	400-00	Nagaraj, Auto no. KA 09 A 8809
17.	Auto fare for the collection of specimen (garden lizard).	No. 17, date 19.01.2014	450-00	Nagaraj, Auto no. KA 09 A 8809
18.	Auto fare for the collection of specimen (garden lizard).	No. 18, date 09.02.2014	600-00	Nataraja, Auto no. KA 09 A 3914
19.	Auto fare for the collection of specimen (garden lizard).	No. 19, date 16.02.2014	500-00	Nataraja, Auto no. KA 09 A 3914
20.	Auto fare for the collection of specimen (garden lizard).	No. 20, date 02.03.2014	400-00	Nataraja, Auto no. KA 09 A 3914
21.	Auto fare for the collection of specimen (garden lizard).	No. 21, date 16.03.2014	450-00	Nataraja, Auto no. KA 09 A 3914

22.	Auto fare for the collection of specimen (garden lizard).	No. 22, date 06.04.2014	500-00	Nataraja, Auto no. KA 09 A 3914
23.	Auto fare for the collection of specimen (garden lizard).	No. 23, date 20.04.2014	350-00	Puttaswamy, Auto no. KA 09 A 2437
24.	Petrol charges for the collection of specimen (garden lizard).	No. 5086, date 11.05.2014	400-00	Shree Siddhi Vinayaka Service Station
25.	Petrol charges for the collection of specimen (garden lizard).	No. 5103, date 25.05.2014	400-00	Shree Siddhi Vinayaka Service Station
26.	Petrol charges for the collection of specimen (garden lizard).	No. 5126, date 08.06.2014	450-00	Shree Siddhi Vinayaka Service Station
27.	Petrol charges for the collection of specimen (garden lizard).	No. 15711, date 22.06.2014	400-00	Trinetra Petroleums
28.	Petrol charges for the collection of specimen (garden lizard).	No. 5166, date 06.07.2014	400-00	Shree Siddhi Vinayaka Service Station
29.	Petrol charges for the collection of specimen (garden lizard).	No. 5190, date 20.07.2014	500-00	Shree Siddhi Vinayaka Service Station
30.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 01878975, date 07.05.2014	70-00	South Western Railway
31.	Train fare from Mysore to Bangalore for the electron microscopy work related to project	No. 06435008, date09.06.2014	30-00	South Western Railway
32.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 08336497, date09.06.2014	55-00	South Western Railway
33.	Bus fare from Majestic Bangalore to Nimhans Bangalore for the electron microscopy work related to project	No. 1624887, date09.06.2014	60-00	South Western Railway
34.	Train fare from Mysore to Bangalore for the electron microscopy work related to project	No. 07672609, date04.08.2014	60-00	South Western Railway
35.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 00379511, date05.08.2014	30-00	South Western Railway
		Total	**13,955-00**	
		Rupees Thirteen Thousand Nine Hundred and Fifty Five only		

BILL NO. 2

Sl. No.	Description of the article	Bill No. and Date	Amount in Rs.	Name of the firm
1.	Auto fare for the collection of specimen (garden lizard).	No. 24, date 03.08.2014	500-00	Nataraja, Auto no. KA 09 A 3914
2.	Auto fare for the collection of specimen (garden lizard).	No. 25, date 17.08.2014	500-00	Nataraja, Auto no. KA 09 A 3914
3.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 00592126 date20.08.2014	225-00	South western railway
4.	Auto fare for the collection of specimen (garden lizard).	No. 26, date 07.09.2014	600-00	Nataraja, Auto no. KA 09 A 3914
5.	Petrol charges for the collection of specimen (garden lizard).	No. 15694, date 17.09.2014	400-00	Trinetra Petroleums
6	Petrol charges for the collection of specimen (garden lizard).	No. 15694, date 02.10.2014	460.00	Trinetra Petroleums
7	Petrol charges for the collection of specimen (garden lizard).	No. 15975, date 08.10.2014	450-00	Trinetra Petroleums
		Total	**3,135-00**	
		Rupees Three Thousand One Hundred and Thirty Five only		

BILL NO. 3

Sl. No.	Description of the article	Bill No. and Date	Amount in Rs.	Name of the firm
1.	Auto fare for the collection of specimen (garden lizard).	No. 26, date 03.11.2014	500-00	Nataraja, Auto no. KA 09 A 3914
2.	Auto fare for the collection of specimen (garden lizard).	No. 27, date 24.11.2014	450-00	Nataraja, Auto no. KA 09 A 3914
3.	Auto fare for the collection of specimen (garden lizard).	No. 28, date 01.12.2014	450-00	Nataraja, Auto no. KA 09 A 3914
4.	Auto fare for the collection of specimen (garden lizard).	No. 2, date 22.12.2014	480-00	Shiva, Auto no. KA 09 A 2437
5.	Auto fare for the collection of specimen (garden lizard).	No. 29, date 05.01.2015	480-00	Shiva, Auto no. KA 09 A 2437
6	Bus fare from Majestic Bangalore to Nimhans Bangalore for the electron microscopy work related to project	No. 1284062, date 17.01.2015	60-00	BMTC
7.	Auto fare for the collection of specimen (garden lizard).	No. 30, date 27.01.2015	500-00	Nataraja, Auto no. KA 09 A 3914
8.	Auto fare for the collection of specimen (garden lizard).	No. 31, date 02.02.2015	450-00	Nataraja, Auto no. KA 09 A 3914
9.	Auto fare for the collection of specimen (garden lizard).	No. 32, date 23.02.2015	500-00	Nataraja, Auto no. KA 09 A 3914
10.	Train fare from Mysore to Davangere for the International conference related to project	No. 4517547115 date 25-02-2015	431-00	South Western Railway (IRCTC)
11.	Auto fare for the collection of specimen (garden lizard).	No. 33, date 02.03.2015	450-00	Nataraja, Auto no. KA 09 A 3914
12.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 50743890, date 12.03.2015	60-00	South Western Railway
13.	Auto fare for the collection of specimen (garden lizard).	No. 34, date 23.03.2015	500-00	Nataraja, Auto no. KA 09 A 3914
14.	Train fare from Mysore to Bangalore for the electron microscopy work related to project	No. 64006493, Date 27.03.2015	60-00	South Western Railway
15.	Train fare from Mysore to Bangalore for the electron microscopy work related to project	No. 50764286, Date 28.03.2015	60-00	South Western Railway
16.	Auto fare for the collection of specimen (garden lizard).	No. 35, date 06.04.2015	500-00	Shiva, Auto no. KA 09 A 2437
17.	Auto fare for the collection of specimen (garden lizard).	No. 36, date 24.04.2015	500-00	Nataraja, Auto no. KA 09 A 3914
18.	Auto fare for the collection of specimen (garden lizard).	No. 37, date 04.05.2015	500-00	Nataraja, Auto no. KA 09 A 3914
19.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 72060387, date 14.05.2015	30-00	South Western Railway
20.	Auto fare for the collection of specimen (garden lizard).	No. 38, date 25.05.2015	500-00	Nataraj, Auto no. KA 09 A 3914
21.	Auto fare for the collection of specimen (garden lizard).	No. 39, date 03.06.2015	500-00	Nataraj, Auto no. KA 09 A 3914
22.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 70951149, date 17.06.2015	30-00	South Western Railway
23.	Train fare from Mysore to Bangalore for the electron microscopy work related to project	No. 66204982 date 17.06.2015	30-00	South Western Railway
24.	Bus fare from Mysore to Manipal for Sperm functionality work in relation to project	No. 36159349 date 22.06.2015	613-00	KSRTC
25.	Auto fare for the collection of specimen (garden lizard).	No. 40, date 25.06.2015	500-00	Nataraja, Auto no. KA 09 A 3914
26.	Bus fare from Manipal to Mysore for Sperm functionality work in relation to project	No. 36159350 date 27.06.2015	553-00	KSRTC
27.	Auto fare for the collection of specimen (garden lizard).	No. 41, date 03.07.2015	500-00	Nataraja, Auto no. KA 09 A 3914

28.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 70788774, date 15.07.2015	60-00	South Western Railway
29.	Petrol charges for the collection of specimen (garden lizard).	No. 22410, date 27.07.2015	470-00	Trinetra petroleums
30.	Petrol charges for the collection of specimen (garden lizard).	No. 22405, date 03.08.2015	450-00	Trinetra petroleums
31.	Petrol charges for the collection of specimen (garden lizard).	No. 22409, date 26.08.2015	450-00	Trinetra petroleums
32.	Petrol charges for the collection of specimen (garden lizard).	No. 22399, date 01.09.2015	450-00	Trinetra petroleums
33.	Train fare from Bangalore to Mysore for the electron microscopy work related to project	No. 89870458, date 23.09.2015	30-00	South Western Railway
34.	Petrol charges for the collection of specimen (garden lizard).	No. 22408, date 24.09.2015	480-00	Trinetra Petroleums
35.	Bus fare from Majestic Bangalore to Nimhans Bangalore for the electron microscopy work related to project	No. 6932322, date 23.09.2015	70-00	BMTC
36.	Auto fare for the collection of specimen (garden lizard).	No. 42, date 02.10.2015	500-00	Nataraja, Auto no. KA 09 A 3914
37.	Bus fare from Mysore to Manipal for Sperm functionality work in relation to project	No. 37907385, date 25.10.2015	678-00	KSRTC
38.	Bus fare from Manipal to Mysore for Sperm functionality work in relation to project	No. 37907386, date 31.10.2015	558-00	KSRTC
		Total	**14,383-00**	
Rupees Fourteen Thousand Three Hundred and Eighty Three only				

BILL NO. 4

Sl. No.	Description of the article	Bill No. and Date	Amount in Rs.	Name of the firm
1.	Auto fare for the collection of specimen (garden lizard).	No. 43, date 05.11.2015	500-00	Nataraja, Auto no. KA 09 A 3914
2.	Auto fare for the collection of specimen (garden lizard).	No. 44, date 03.12.2015	450-00	Nataraja, Auto no. KA 09 A 3914
3.	Auto fare for the collection of specimen (garden lizard).	No. 45, date 24.12.2015	450-00	Nataraja, Auto no. KA 09 A 3914
4.	Auto fare for the collection of specimen (garden lizard).	No. 46, date 04.01.2016	480-00	Shiva, Auto no. KA 09 A 2437
5.	Auto fare for the collection of specimen (garden lizard).	No. 47, date 26.01.2016	480-00	Shiva, Auto no. KA 09 A 2437
6	Auto fare for the collection of specimen (garden lizard).	No. 48, date 04.02.2016	500-00	Nataraja, Auto no. KA 09 A 3914
7	Auto fare for the collection of specimen (garden lizard).	No. 49, date 27.02.2016	450-00	Nataraja, Auto no. KA 09 A 3914
8	Auto fare for the collection of specimen (garden lizard).	No. 50, date 03.03.2016	220-00	Nataraja, Auto no. KA 09 A 3914
		Total	**3,530-00**	
Rupees Three Thousand Five Hundred and Thirty only				

Condensed travel bills from the above

Name of the Article	Duration of Travel/Field Work		Expenditure Incurred (Rs.)
Bill No. 1 (from above)	From 19-07-2013	To 05-08-2014	**13, 955/-**
Bill No. 2 (from above)	From 03-08-2014	To 08-10-2014	**3, 135/-**
Bill No. 3 (from above)	From 03-11-2014	To 31-10-2015	**14, 383/-**
Bill No. 4 (from above)	From 05-11-2015	To 03-03-2016	**3, 530/-**
		Total	**35,003/-**
Rupees Thirty Five thousand and Three only			

Certified that the above expenditure is in accordance with the UGC norms for Major Research Projects.

PRINCIPAL INVESTIGATOR

Dr. T. RUTH SHANTHAKUMARI
Principal Investigator-MRP
Dept. of Zoology
St. Philomena's College, Mysore-570 015

PRINCIPAL

Principal
St. Philomena's College
MYSORE-570 015

PROFORMA FOR SUPPLYING THE INFORMATION IN RESPECT OF THE STAFF APPOINTED UNDER THE SCHEME OF MAJOR RESEARCH PROJECT

UGC File No. F. 42-615/2013(SR) dated 25th March 2013

Title of the project **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*"**

1.	Name of the Principal Investigator	Dr. Ruth Shanthakumari T			
2.	Name of the Department/University	Dept. of Zoology,St. Philomena's College, Bannimantap, Mysore- 570006			
3.	Name of the Research Personnel appointed	Mr. Goutham S			
4.	Academic qualification	S. No	Qualification	Year	Marks %
		1.	M.Sc	2010	66.75
		2.	Project assistant	07-05-2013 to 31-03-2017	-
		3.	Ph.D	Enrolled on 11-09-2013	-
5.	Date of joining	07-05-2013			
6.	Date of birth of the research personnel	03-10-1985			
7.	Amount of HRA if drawn	Nil			
8.	Number of candidates applied for the post	4			

Certificate

This is to certify that all the rules and regulations of the UGC Major Research Project outlined in the guidelines have been followed. Any lapse on the part of the University will be liable to terminate of the said UGC project.

Principal Investigator
Dr. T. RUTH SHANTHAKUMARI
Principal Investigator-MRP
Dept. of Zoology
St. Philomena's College, Mysore-570 015

Head of Department
CECILIA D'SOUZA, M.Sc.
Head of Department of Zoology
St. Philomena's College
Mysore-570 015

Principal
Principal
St. Philomena's College
MYSORE-570 015

Annexure –VIII

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

FINAL REPORT OF THE WORK DONE ON THE MAJOR RESEARCH PROJECT

1. Project report number **Final**
2. UGC Reference No. & Date No. F. 42-615/2013(SR) dated 25th March 2013
3. Period of report 07-05-2013 to 31-03-2017
4. Title of research project **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*"**
5. (a) Name of the Principal Investigator : **Dr. Ruth Shanthakumari T**

   (b) Department Department of Zoology

   (c) University/ College where the project was undertaken St. Philomena's College, Mysore
6. Effective date of starting of the project: 07-05-2013
7. Grant approved and expenditure incurred during the period of the report:
   a) Total amount approved Rs: **8, 35, 800/-**
   b) **Total amount allocated Rs: 8,16,703/-**
   c) Total amount released Rs: **7, 42, 312/-**
   d) Total expenditure Rs. **7, 39, 353/-**
   e) Total amount to be release Rs: **74, 391/-**
   f) **Final Total expenditure: 8,13,593/-**
   g) Report of the work done

i.	Objectives of the Project	1. Investigation on histochemical and ultrastructural details of sperm storage pockets 2. Analysis of pro-survival factors, in flushings of sperm storage pockets using classical biochemical techniques 3. Testing the efficacy of the pro-survival factor(s), in *in vivo* and/or *in vitro* condition
ii.	Work done so far	➢ Histological and histochemical features of sperm storage pockets ➢ Ultrastructural features of sperm storage tubules. ➢ Purified of suspected sperm motility inhibitor protein/s from the uterovaginal flushings ➢ Tested the efficacy of the sperm motility inhibitor protein/s of the uterovaginal flushings on washed epididymal sperms. (Details are mentioned in the progress report)

	Publications, if any, resulting from the work	➢ Shankar G, Sagar BKC, Kumari TRS and Marathe GK. (2015). Ultrastructural features of sperm storage tubules in the oviduct of the Indian garden lizard, *Calotes versicolor*. Anal Rec. 298:1932–1937. (copy attached)
iii.	Has the progress been according to original plan of work and towards achieving objectives if not, state reasons	Yes
iv.	Please indicate the difficulties, if any, experienced in implementing the project	Second installment of the grant was not released in time.
v.	If project has not been completed, please indicate the approximate time by which it is likely to be completed.	Project has been completed
vi.	If the project has been completed, please enclose a summary of the findings of the study. One bound copy of the final report of work done may also be sent to University Grants Commission.	A summary of the work done for the period (Annual basis) may please be sent to the Commission on a separate sheet.
vii.	Any other information which would help in evaluation of work done on the project. At the completion of the project, the first report should indicate the output, such as (a) Manpower trained (b) Ph. D. awarded (c) Publication of results (d) other impact, if any	a) Manpower trained: 3 M. Sc Students carried out their project work on the topics related to the MRP and were trained in the project. b) PhD: One candidate registered for Ph.D (enrolment c) No.WF-1349/2013-14 dt. 11-09-2013). d) Publications: 1 papers published e) Other impact: Findings from this project will be utilized as preliminary data by PI for further research in the area and in writing next grant(s).

PRINCIPAL INVESTIGATOR
Dr. T. RUTH SHANTHAKUMARI
Principal Investigator-MRP
Dept. of Zoology
St. Philomena's College, Mysore-570 015

PRINCIPAL
Principal
St. Philomena's College
MYSORE-570 015

Annexure – IX

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

PROFORMA FOR SUBMISSION OF INFORMATION AT THE TIME OF SENDING THE FINAL REPORT OF THE WORK DONE ON THE PROJECT

1. **TITLE OF THE PROJECT**: **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*"**
2. **NAME AND ADDRESS OF THE PRINCIPAL INVESTIGATOR**:

   Dr. Ruth Shanthakumari T,

   #609, 'MELINMANE'

   9th Cross, Roopanagar,

   Mysuru-570026

   Karnataka, India.
3. **NAME AND ADDRESS OF THE INSTITUTION**:

   Department of Zoology,

   St. Philomenas' College, Bannimantap,

   Mysuru-570015,

   Karnataka, India.
4. **UGC APPROVAL LETTER NO. AND DATE**: No. F. 42-615/2013(SR) dated 25th March 2013
5. **DATE OF IMPLEMENTATION: 07-05-2013**
6. **TENURE OF THE PROJECT: 3 years+1 year extension (25-03-2013 to 31-03-2017)**
7. **TOTAL GRANT ALLOCATED: Rs. 8, 16, 703/-**
8. **TOTAL GRANT RECEIVED: RS. 7, 42, 312/-**
9. **TOTAL AMOUNT RELEASED RS: 7, 42, 312/-**
10. **FINAL EXPENDITURE: RS. 7, 39, 353 +74,240 = 8,13,593/-**
11. **TOTAL AMOUNT TO BE RELEASED RS: 74, 391/-**
12. **TITLE OF THE PROJECT: "Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*"**

13. OBJECTIVES OF THE PROJECT:

1. Investigation on histochemical and ultrastructural details of sperm storage pockets.
2. Analysis of pro-survival factors, in flushings of sperm storage pockets using classical biochemical techniques.
3. Testing the efficacy of the pro-survival factor(s), in *in vivo* and/or *in vitro* condition

14. WHETHER OBJECTIVES WERE ACHIEVED:

Yes.

The first objective of the project histological and histochemical study was carried out in St. Philomena's college, Mysuru. The ultrastructural studies were carried out in National Institute of Mental Health and Neurosciences (NIHANS) Bengaluru.

The second objective was to purify the suspected sperm motility inhibitor protein/s from the uterovaginal flushings using classical chromatography techniques, which were carried out in the University of Mysore, Mysuru.

The third objective was to check the effect of partially purified epididymal sperm motility inhibitor protein/s from the uterovaginal flushings, which was successfully carried out in Clinical Embryology Laboratory, Kasturba Medical College, Manipal and University of Mysore, Mysuru.

15. ACHIEVEMENTS FROM THE PROJECT:

- The current study provides the first depiction of the ultrastructural features of SST during three phases of the reproductive cycle in *C. versicolor*, which showed cyclical changes in the secretory activity of the epithelial cells lining the SST.
- In this project were able to recover live motile sperm from the uterovaginal region of the oviduct of *C. versicolor*
- Based on the above study we were able to publish the work entitled :Ultrastructural features of sperm storage tubules in the oviduct of the Indian garden lizard, *Calotes versicolor*" in an International journal –The Anatomical record
- We also observed that in a large majority of the female lizards isolated from males after mating far up to 70 days, the uterovaginal flushings still contained live motile sperm, which provide circumstantial evidence for sperm storage.
- Most importantly we purified a ~55kDa abundant protein from the uterovaginal flushings that lacked homology to any protein on the NCBI database, this protein reversibly inhibited the motility of washed epididymal sperm of *C. versicolor* in a concentration and time-dependent fashion.
- We are currently writing a manuscript on the effect of ~55kDa protein of uterovaginal flushing on sperm parameters.
- Based on the interesting results obtained in the current project we are planning to apply for higher grants to further continue the work, where we want to look at the effect of this protein on

sperm of other animals and also on the effect of peptide fragments resulting from protease digestion from this protein on sperm motility.

16. SUMMARY OF THE FINDINGS:

Sperm storage can be defined as the maintenance of sperm inside a female's reproductive tract for an extended period of time in a viable condition. Sperm storage in the female reproductive tract has been reported in a variety of animals including insects, fishes, amphibians, reptiles, birds, and mammals. With respect to the mechanism involved in the sperm storage, not much has been known. A foreign cell being able to survive in viable condition inside the reproductive tract is a very intriguing feature attracting a lot of attention.

This study provides the first description of the ultrastructural features of sperm storage tubules (SSTs) in the uterovaginal region of the oviduct of the Indian garden lizard, *C. versicolor*. The present study is also the first report of recovery of motile sperm from the flushings of SSTs in *C. versicolor*. Partially purified single abundant protein of the uterovaginal flushings was found to be a unique protein with not having any homology with known protein in the NCBI database. Further this partially purified protein was able to retard the motility without affecting much of the viability of the washed epididymal sperms *in vitro*.

CONTRIBUTION TO THE SOCIETY:

- Establishment of reptilian model for the study of sperm storage.
- Our study strongly indicates the presence of sperm motility inhibitor protein/s in the uterovaginal flushings. Further research can be carried out to understand the effect of this protein on sperm of heterologous species.
- Training both Ph.D and M Sc students to take up the research problems

17. WHETHER ANY PH.D. ENROLLED/PRODUCED OUT OF THE PROJECT:

Yes. Mr. Goutham S, project fellow working as full time research scholar, registered for Ph.D (WF-1349/2013-14 dt. 11/09/3013).

Thesis Title: "Unraveling the biology of sperm storage in the oviduct of the Indian garden lizard, *Calotes versicolor*".

18. NO. OF PUBLICATIONS OUT OF THE PROJECT: 1, (copy attached).

Shankar G, Sagar BKC, Kumari TRS and Marathe GK. (2015). Ultrastrructural features of sperm storage tubules in the oviduct of the Indian garden lizard, Calotes versicolor. Anal Rec. 298:1932–1937.

PRINCIPAL INVESTIGATOR
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Annexure – X

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

ASSESSMENT CERTIFICATE

It is certified that the proposal entitled **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*".** by **Dr. Ruth Shanthakumari T** Dept. of Zoology has been assessed by the two member committee consisting the following members for submission to the University Grants Commission, New Delhi for financial support under the scheme of Major Research Projects:

Details of Expert Committee:

1. **Cecilia D' Souza,**
   Head of the Department,
   Department of Zoology,
   St. Philomena's College,
   Bannimantap,
   Mysuru-570015

   Cecilia D'souza
   CECILIA D'SOUZA, M.Sc.
   Head of Department of Zoology
   St. Philomena's College
   Mysore-570 015

2. **Dr. Gopal Marathe K,**
   Associate Professor,
   Department of Studies in Biochemistry,
   University of Mysore,
   Mysuru-570006

   Dr. GOPAL MARATHE. K. MSc.,Ph.D.,
   Associate Professor
   Dept. of Studies in Biochemistry
   University of Mysore, Manasagangotri
   Mysore-570006

The proposal is as per the guidelines.

PRINCIPAL

Principal
St. Philomena's College
MYSORE-570 015

Annexure – X

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

ASSESSMENT CERTIFICATE

It is certified that the proposal entitled **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*".** by **Dr. Ruth Shanthakumari T** Dept. of Zoology has been assessed by the three member committee consisting the following members for submission to the University Grants Commission, New Delhi for financial support under the scheme of Major Research Projects:

Details of Expert Committee:

1. **Cecilia D' Souza,**
Head of the Department,
Department of Zoology,
St. Philomena's College,
Bannimantap,
Mysuru-570015

CECILIA D'SOUZA, M.Sc.
Head of Department of Zoology
St. Philomena's College
Mysore-570 015

2. **Dr. Gopal Marathe K.,**
Associate Professor,
Department of Studies in Biochemistry,
University of Mysore,
Mysuru-570006

Dr. GOPAL MARATHE. K. MSc.,Ph.D.
Associate Professor
Dept. of Studies in Biochemistry
University of Mysore, Manasagangotri
Mysore-570006

2. **Prof. Kemparaju K.,**
Department of Studies in Biochemistry,
University of Mysore,
Mysuru-570006

Chairman
Dept. of Studies in Biochemistry
University of Mysore
Manasagangotri, MYSORE-570 006
INDIA

The proposal is as per the guidelines.

PRINCIPAL

Principal
St. Philomena's College
MYSORE-570 015

Annexure – XI

Final Report Assessment / Evaluation Certificate

(Two Members Expert Committee Not Belonging to the Institute of Principal Investigator)

It is certified that the final report of Major Research Project **(F. No. 42-615/2013 (SR))** entitled "**Unraveling the biology of sperm storage in the oviduct of the female Indian garden lizard, *Calotes versicolor***" by, **Dr. T. Ruth Shantha Kumari,** Dept. of Zoology, St. Philomena's College Mysuru, has been assessed by the committee consisting the following members for final submission of the report to the UGC, New Delhi under the scheme of Major Research Project.

Comments/Suggestions of the Expert Committee:-

The work has been satisfactorily carried out with two outstanding publications and has the potential to apply for a new proposal.

Name & Signatures of Experts with Date:-

Name of Expert	University/College name	Signature with Date
1. Dr. Gopal Marathe K.	Dept. of Studies Biochemistry University of Mysore, Manasagangotri, Mysuru, Karnataka, India	04/05/2018 Dr. GOPAL MARATHE K. MSc.,Ph.D. Associate Professor Dept. of Studies in Biochemistry University of Mysore, Manasagangotri Mysore-570006
2. Prof. Kemparaju K.	Dept. of Studies Biochemistry University of Mysore, Manasagangotri, Mysuru, Karnataka, India	04/5/2018 Chairman Dept. of Studies in Biochemistry University of Mysore Manasagangotri, MYSORE-570 006 INDIA

It is also certified that final report, Executive summary of the report, Research documents, monograph academic papers provided under Major Research Project have been posted on the website of the College.

(Principal)

Principal
St. Philomena's College,
MYSORE - 570 015
Karnataka, India.

Final Report / Executive Summary of UGC-major research project, entitled "Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*" submitted by the Institution

Sperm storage can be defined as the maintenance of sperm inside a female's reproductive tract for an extended period of time in a viable condition. Sperm storage in the female reproductive tract has been reported in a variety of animals including insects, fishes, amphibians, reptiles, birds, and mammals. With respect to the mechanism involved in the sperm storage, not much has been known. A foreign cell being able to survive in viable condition inside the reproductive tract is a very intriguing feature attracting a lot of attention.

In the present UGC-Major Research Project using the Indian garden lizard, *Calotes versicolor* as a model to unravel the biology of sperm storage the following research findings were made which are summarized as follows.

- The current study provides the first depiction of the ultrastructural features of SST during three phases of the reproductive cycle in *C. versicolor*, which showed cyclical changes in the secretory activity of the epithelial cells lining the SST.
- Live motile sperm were recovered from the uterovaginal region of the oviduct of *C. versicolor* for an extended
- The above research finding resulted in an International publication in the journal –The Anatomical record, entitled "Ultrastructural features of sperm storage tubules in the oviduct of the Indian garden lizard, *Calotes versicolor*"
- In a large majority of the female lizards isolated from males after mating far up to 70 days, the uterovaginal flushings still contained live motile sperm, which provide circumstantial evidence for sperm storage.
- An abundant ~55kDa protein was purified from the uterovaginal flushings that lacked homology to any protein on the NCBI database, this protein reversibly inhibited the motility of washed epididymal sperm of *C. versicolor* in a concentration and time-dependent fashion.
- The effect of ~55kDa protein of uterovaginal flushings on sperm parameters resulted one more International publication "Proteinaceous sperm motility inhibitory factor from the female Indian garden lizard, *Calotes versicolor*" in the journal Reproduction, Fertility and Development.
- The current project provides a platform for further research, where one can look at the molecular mechanism(s) involved in reversible inhibition of sperm motility by ~55-kDa protein

Principal
St. Philomena's College,
MYSORE - 570 015
Karnataka, India.

Final Report / Executive Summary Certificate

It is certified that the final report / Executive Summary of Major Research Project **(F. No. 42-615/2013 (SR))** entitled "**Unraveling the biology of sperm storage in the oviduct of the female Indian garden lizard, *Calotes versicolor***" by, **Dr. T. Ruth Shantha Kumari,** Dept. of Zoology, St. Philomena's College Mysuru., has been placed in the Institution Website and Kept in Library for final submission of the report as per the UGC, New Delhi under the scheme of Major Research Project.

(Principal)

Principal
St. Philomena's College,
MYSORE - 570 015
Karnataka, India.

Annexure – V

UNIVERSITY GRANTS COMMISSION

BAHADUR SHAH ZAFAR MARG

NEW DELHI – 110 002

Utilization certificate

Certified that the grant of Rs. **7, 42, 312/-** (Rupees Seven lakh Forty Two thousand Three hundred and Twelve only) received from the University Grants Commission in **two installments (1st installment of Rs. 4,54,300 and 2nd installment of Rs. 2,88,012)** under the scheme of support for Major Research Project entitled **"Unraveling the biology of sperm storage in the oviduct of female garden lizard, *Calotes versicolor*"** vide UGC letter No. F. 42-615/2013(SR) dated 25th March 2013 **(I-grant)** and dated 29th Aug 2015 **(II-grant). Rupees Seven lakh Thirty Nine thousand Three hundred and Fifty Three (7, 39, 353/-)** has been utilized for the purpose for which it was sanctioned and in accordance with the terms and conditions laid down by the University Grants Commission.

Amount of **74,240/- (Rupees Seventy Four thousand Two hundred and Forty)** [Honorarium: 50,666/-, Chemical/Glassware/Consumable: 15,021/-, Travel/Field work: 3,530/- and Contingency: 5,023/- (10% of the total grant amount which has been withheld by University Grants Commission)] under recurring charges has been spent by the institute in advance for the purpose of the project, and will be reimbursed by the institute once it is released by the UGC-MRP.

HARIHARAN & CO. 4581 'GOMATHI' N.R. Mohalla Mysore-7 CHARTERED ACCOUNTANTS

PRINCIPAL INVESTIGATOR

Dr. T. RUTH SHANTHAKUMARI
Principal Investigator-MRP
Dept. of Zoology
St. Philomena's College, Mysore-570 015

STATUTORY AUDITOR

FOR HARIHARAN & CO.
Chartered Accountants
Firm Regn. No. 001084S
(M. PERUMAL)
Partner
Mem. No. 200165

REGISTRAR/PRINCIPAL

Principal
St. Philomena's College
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